Content-Based Image Retrieval (CBIR) for Chest X-rays Using Histogram, GLCM, and Statistical Features

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***Abstract*—This mid-semester report presents the progress made in developing a Content-Based Image Retrieval (CBIR) system for chest X-ray images using handcrafted features. The system is based on grayscale histogram features, Gray Level Co-occurrence Matrix (GLCM) texture descriptors, Local Binary Patterns (LBP), statistical descriptors, and Euclidean distance as the similarity metric. Image preprocessing, feature extraction, feature caching, and initial retrieval experiments have been successfully implemented. This document summarizes the methodology, implementation progress, and results achieved up to the midterm stage.**

***Index Terms*—CBIR, Chest X-ray, Histogram, GLCM, LBP, Texture Features, Euclidean Distance, Image Retrieval**

I. Introduction

Content-Based Image Retrieval (CBIR) is an image search technique that retrieves images based on intrinsic visual features rather than metadata or textual tags. In medical imaging, CBIR helps doctors find similar cases, compare abnormalities, and support diagnosis. Chest X-rays contain important visual patterns—texture, intensity gradients, and structural differences—making them a suitable domain for CBIR.

This project aims to design a CBIR system that extracts visual characteristics from chest X-ray images using histogram, GLCM, statistical measures, and LBP features. These features are then compared using Euclidean distance to retrieve the most visually similar images from a database.

II. Problem Definition

Medical databases contain thousands of chest X-ray images. Manually searching for similar cases is time-consuming and prone to human error. An automated system is needed to retrieve images that resemble a given query image based on pixel patterns and texture.

The goal is to build a CBIR system that:

- Extracts meaningful image features (intensity + texture)
- Stores feature vectors efficiently
- Measures similarity using statistical distance metrics
- Retrieves and ranks the top similar X-ray images

III. Objectives

- Build a preprocessing pipeline for Chest X-ray images.
- Extract Histogram, GLCM, LBP, and statistical features.
- Implement Euclidean distance–based similarity measurement.
- Create a feature database for fast retrieval.
- Retrieve and rank the top-K similar images.
- Prepare for performance evaluation using precision/recall.

IV. Literature Review

Several research works inspired the feature selection process:

A. Texture-Based X-ray Retrieval

Ganesan and Subashini [1] demonstrated that texture descriptors like GLCM are effective for distinguishing patterns in X-ray images. Their approach influenced our use of multi-angle GLCM.

B. Statistical Texture Descriptors

Shivamurthy and Manjunatha [2] proposed statistical texture-based CBIR for chest X-rays, highlighting that features such as correlation and homogeneity improve retrieval quality.

C. Shape Histogram and Global Descriptors

Azevedo-Marques [3] showed that histogram-based global descriptors support image classification when combined with texture features.

These works collectively support our feature extraction pipeline.

V. Dataset

The NIH ChestX-ray14 dataset from Kaggle is used.

- **Source:** National Institutes of Health (NIH)
- **Format:** PNG, JPEG
- **Total Images:** 112,000+ (we used 10–15 for initial tests)
- **Diseases:** Pneumonia, Effusion, Cardiomegaly, etc.

VI. Methodology

This section describes the complete workflow used in our CBIR system, including image preprocessing, feature extraction, feature storage, and similarity-based retrieval.

A. Image Preprocessing

Each X-ray undergoes:

- Grayscale conversion
- Resizing to 256×256
- Normalization
- CLAHE contrast enhancement

B. Feature Extraction

We extract multiple handcrafted features:

1) Histogram Features: Global intensity distribution using 64 bins.

2) GLCM Features: Computed using four orientations: 0°, 45°, 90°, 135°.

3) Local Binary Pattern (LBP): Uses uniform patterns with radius 1 and 8 neighbors.

4) Statistical Features: Includes mean, variance, standard deviation.

C. Feature Caching and Parallelization

All extracted features are saved into NumPy arrays for faster retrieval.

D. Similarity Metric

Euclidean distance is used to rank similar images.

E. Rank Retrieval

Top-K most similar images are returned.

VII. Implementation Code

```python
import cv2
import numpy as np
from skimage.feature import graycomatrix, graycoprops
from sklearn.preprocessing import normalize

# Preprocess image: grayscale + resize + CLAHE
def preprocess_image(path):
    img = cv2.imread(path, cv2.IMREAD_GRAYSCALE)
    img = cv2.resize(img, (256, 256))
    clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
    return clahe.apply(img)

# Histogram feature (64 bins)
def extract_histogram(img):
    h = cv2.calcHist([img],[0],None,[64],[0,256]).flatten()
    return h / (h.sum() + 1e-9)

# GLCM texture features
def extract_glcm(img):
    glcm = graycomatrix(img, [1],
           [0, np.pi/4, np.pi/2, 3*np.pi/4],
           levels=256, symmetric=True, normed=True)
    props = ['contrast','correlation','energy','homogeneity']
    return np.array([graycoprops(glcm,p).mean() for p in props])
```

Fig. 1. Query image and its histogram (left column).

```python

# Combined feature vector
def extract_features(path):
    img = preprocess_image(path)
    hist = extract_histogram(img)
    glcm = extract_glcm(img)
    return np.concatenate([hist, glcm])

# Euclidean distance retrieval
def retrieve_similar(query_feat, db_feats, paths, k=5):
    d = np.linalg.norm(db_feats - query_feat, axis=1)
    idx = np.argsort(d)[:k]
    return [(paths[i], d[i]) for i in idx]
```

VIII. Progress So Far

- Completed image preprocessing pipeline.
- Implemented Histogram, GLCM, LBP, and Statistical features.
- Implemented Euclidean similarity function.
- Integrated multi-threaded feature extraction.
- Built retrieval function that displays query + Top-5 similar images.
- Validated functionality with subset of dataset.

IX. Results

The histogram results in four figures that show the grayscale intensity distributions for four chest X-ray samples. Each subfigure includes the original image and its normalized histogram, highlighting variations in brightness and contrast across the dataset. Sample (a) displays a strong low-intensity peak, indicating a darker image, whereas Sample (b) shows a broader mid-intensity spread consistent with higher contrast lung regions. Sample (c) presents a smoother, more uniform histogram, suggesting fewer abrupt intensity changes, while Sample (d) exhibits wider fluctuations, reflecting greater variation in tissue shading. These differences demonstrate that histograms effectively capture global intensity characteristics and provide a useful basis for distinguishing X-ray images within the CBIR system.

A. Histogram Feature Results

Histogram features capture the global intensity distribution of chest X-ray images. Images with similar brightness and contrast patterns tend to have similar histogram shapes. Below, we compare multiple X-ray images along with their normalized histograms. The peaks and slopes in the histogram represent distribution of grayscale intensities, which helps the CBIR system understand overall brightness patterns, lung texture lightness, and contrast differences across images.

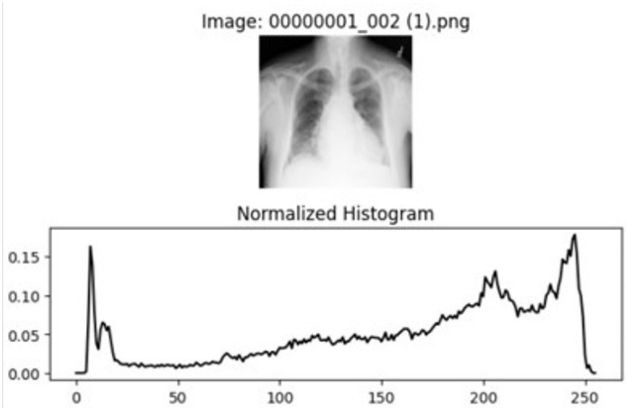

Fig. 2. *
(a) Sample X-ray + Histogram (1)

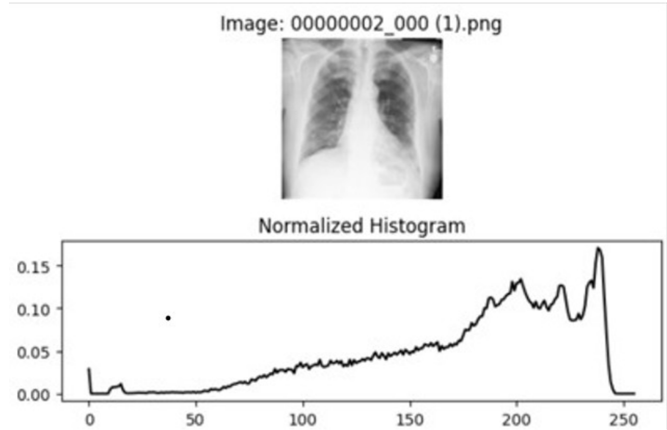

Fig. 3. *
(b) Sample X-ray + Histogram (2)

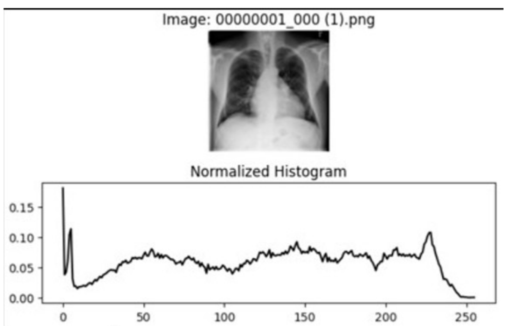

Fig. 4. *
(c) Sample X-ray + Histogram (3)

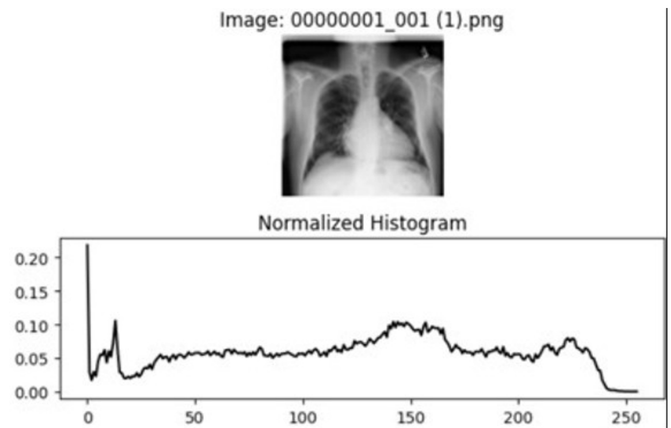

Fig. 5. *
(d) Sample X-ray + Histogram (4)

Fig. 6. Comparison of sample X-ray images and normalized histograms.

X. Plan for Next Phase

- Implement Cosine, Chi-square, Manhattan distance.
- Evaluate with Precision@K and Recall.
- Expand dataset size.

XI. References

References

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